Short-Kappa Branch of Maximal-Parabolic Induction

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1 lemma lem:x0-geometry-and-k

1.1 statement

One has

$$(\operatorname{Im} X_0)^{\perp} = \ker X_0.$$

Hence the restriction of $\langle \cdot, \cdot \rangle_0$ to $\ker X_0$ has radical

$$R := \ker X_0 \cap \operatorname{Im} X_0,$$

and therefore induces a non-degenerate ϵ-symmetric form on

$$K_{X_0} := \ker X_0 / R.$$

Moreover there exists a subspace

$$V_1 \subset \ker X_0$$

such that the quotient map

$$q : \ker X_0 \to K_{X_0}$$

restricts to an isometric isomorphism

$$q|_{V_1} : V_1 \xrightarrow{\sim} K_{X_0}.$$

For such a choice one has an orthogonal decomposition

$$\ker X_0 = V_1 \oplus R$$

and

$$V_0 = V_1 \oplus V_1^{\perp},$$

where $V_1^{\perp}$ is X_0-stable.

Finally, if $h \in \operatorname{Isom}(K_{X_0}, \langle \cdot, \cdot \rangle_K)$, then there is an element

$$\tilde{h} \in Z_{G_0}(X_0)$$

whose restriction to V_1 is $(q|_{V_1})^{-1} h (q|_{V_1})$ and which acts trivially on $V_1^{\perp}$.

1.2 proof

If $w \in \ker X_0$ and $y = X_0 z \in \operatorname{Im} X_0$, then

$$\langle w, y\rangle_0 = \langle w, X_0 z\rangle_0 = -\langle X_0 w, z\rangle_0 = 0.$$

So

$$\ker X_0 \subseteq (\operatorname{Im} X_0)^{\perp}.$$

Since

$$\dim(\operatorname{Im} X_0)^{\perp} = \dim V_0 - \dim \operatorname{Im} X_0 = \dim \ker X_0,$$

the inclusion is an equality.

Therefore the radical of the restriction of $\langle \cdot, \cdot\rangle_0$ to $\ker X_0$ is

$$\ker X_0 \cap (\ker X_0)^{\perp} = \ker X_0 \cap \operatorname{Im} X_0 = R,$$

so the form descends to a non-degenerate ϵ-symmetric form on K_{X_0}.

Choose any linear section $s : K_{X_0} \to \ker X_0$ of q. The pulled-back form

$$(a, b) \longmapsto \langle s(a), s(b)\rangle_0$$

is exactly $\langle a, b\rangle_K$, hence is non-degenerate. Therefore the image

$$V_1 := s(K_{X_0})$$

is a non-degenerate subspace of $\ker X_0$, $q|_{V_1}$ is an isometric isomorphism, and

$$\ker X_0 = V_1 \oplus R$$

orthogonally.

Because V_1 is non-degenerate in V_0, we have

$$V_0 = V_1 \oplus V_1^{\perp}.$$

If $v \in V_1^{\perp}$ and $u \in V_1$, then $X_0 u = 0$, so

$$\langle X_0 v, u\rangle_0 = -\langle v, X_0 u\rangle_0 = 0.$$

Thus $X_0 v \in V_1^{\perp}$, proving that $V_1^{\perp}$ is X_0-stable.

Now let $h \in \operatorname{Isom}(K_{X_0})$. Via the isometry $q|_{V_1} : V_1 \to K_{X_0}$, it defines an isometry of V_1, still denoted h. Extend it to V_0 by letting it act trivially on $V_1^{\perp}$. Since $V_0 = V_1 \oplus V_1^{\perp}$ is orthogonal, this extension is an element $\tilde{h} \in G_0$.

Because X_0 vanishes on V_1 and preserves $V_1^{\perp}$, and because $\tilde{h}$ acts as the identity on $V_1^{\perp}$, we have

$$\tilde{h} X_0 = X_0 \tilde{h}.$$

So $\tilde{h} \in Z_{G_0}(X_0)$, as required.

2 lemma lem:parametrize-x0-plus-u

2.1 statement

For $C \in \mathrm{Hom}_F(E', V_0)$ let $C^\vee : V_0 \to E$ be defined by

$$\lambda(C^\vee v, e') = -\langle v, Ce' \rangle_0.$$

For $B \in \mathrm{Hom}_F(E', E)$ satisfying

$$\lambda(Bu, v) + \epsilon\lambda(Bv, u) = 0 \qquad (u, v \in E'),$$

define

$$X_{C,B}(e + v + e') := C^\vee v + Be' + X_0 v + Ce'.$$

Then $X_{C,B} \in X_0 + \mathfrak{u}$.

Conversely, every element of $X_0 + \mathfrak{u}$ is uniquely of the form $X_{C,B}$.

2.2 proof

Relative to the decomposition $V = E \oplus V_0 \oplus E'$, an element $Y \in \mathfrak{u}$ has the shape

$$Y = \begin{pmatrix} 0 & A & B \\ 0 & 0 & C \\ 0 & 0 & 0 \end{pmatrix}$$

with $A : V_0 \to E$, $B : E' \to E$, $C : E' \to V_0$. The condition $Y \in \mathfrak{g}$ is

$$\langle Yx, y \rangle + \langle x, Yy \rangle = 0.$$

Testing this on $(v, e') \in V_0 \times E'$ gives

$$\lambda(Av, e') + \langle v, Ce' \rangle_0 = 0,$$

so $A = C^\vee$. Testing it on $(u, v) \in E' \times E'$ gives

$$\lambda(Bu, v) + \epsilon\lambda(Bv, u) = 0.$$

Hence every $Y \in \mathfrak{u}$ is uniquely of the form

$$Y_{C,B}(e + v + e') = C^\vee v + Be' + Ce'.$$

Therefore every element of $X_0 + \mathfrak{u}$ is uniquely

$$X_0 + Y_{C,B} = X_{C,B}.$$

3 lemma lem:unipotent-conjugation

3.1 statement

For $D \in \mathrm{Hom}_F(E', V_0)$ define

$$u_D(e + v + e') := \left(e + D^\vee v + \frac{1}{2} D^\vee D\, e'\right) + (v + De') + e'.$$

Then $u_D \in P$. Moreover, for every pair (C, B) as above,

$$u_D X_{C,B} u_D^{-1} = X_{C - X_0 D,\; B + D^\vee C - C^\vee D - D^\vee X_0 D}.$$

3.2 proof

In block form,

$$u_D = \begin{pmatrix} 1 & D^\vee & \frac{1}{2}D^\vee D \\ 0 & 1 & D \\ 0 & 0 & 1 \end{pmatrix},$$

and its inverse is

$$u_D^{-1} = \begin{pmatrix} 1 & -D^\vee & \frac{1}{2}D^\vee D \\ 0 & 1 & -D \\ 0 & 0 & 1 \end{pmatrix}.$$

To check that u_D preserves the form, it is enough to test it on pairs involving the E'-summand.

For $v \in V_0$ and $e' \in E'$,

$$\langle D^\vee v, e' \rangle = \lambda(D^\vee v, e') = -\langle v, De' \rangle_0,$$

so the mixed (V_0, E')-terms cancel.

For $u', v' \in E'$,

$$\begin{aligned} \langle u_D u', u_D v' \rangle &= \lambda\Big(\frac{1}{2}D^\vee D\, u', v'\Big) + \epsilon\lambda\Big(\frac{1}{2}D^\vee D\, v', u'\Big) + \langle Du', Dv' \rangle_0 \\ &= -\frac{1}{2}\langle Du', Dv' \rangle_0 - \frac{\epsilon}{2}\langle Dv', Du' \rangle_0 + \langle Du', Dv' \rangle_0 \\ &= 0, \end{aligned}$$

because $\langle Dv', Du' \rangle_0 = \epsilon \langle Du', Dv' \rangle_0$. Hence $u_D \in G$, and since it fixes E and stabilizes $E^\perp$, in fact $u_D \in P$.

Now

$$X_{C,B} = \begin{pmatrix} 0 & C^\vee & B \\ 0 & X_0 & C \\ 0 & 0 & 0 \end{pmatrix}.$$

Multiplying the three block matrices u_D, $X_{C,B}$, and u_D^{-1} gives

$$u_D X_{C,B} u_D^{-1} = \begin{pmatrix} 0 & C^\vee + D^\vee X_0 & B + D^\vee C - C^\vee D - D^\vee X_0 D \\ 0 & X_0 & C - X_0 D \\ 0 & 0 & 0 \end{pmatrix}.$$

Since

$$\big(C - X_0 D\big)^\vee = C^\vee + D^\vee X_0$$

by the defining relation for $(-)^\vee$, this is exactly

$$X_{C - X_0 D,\, B + D^\vee C - C^\vee D - D^\vee X_0 D}.$$

This proves the formula.

4 proposition prop:generic-normal-form

4.1 statement

Fix a d-dimensional subspace

$$A \subset V_1 \simeq K_{X_0}.$$

Let

$$W_A \subset V_0 / \operatorname{Im} X_0$$

be the annihilator of A under the perfect pairing

$$(V_0 / \operatorname{Im} X_0) \times \ker X_0 \to F, \qquad (\overline{v}, k) \mapsto \langle k, v \rangle_0.$$

Then $\dim W_A = \kappa$.

Choose any κ-dimensional subspace $V_A \subset V_0$ mapping isomorphically to W_A, and any isomorphism

$$S_A : E' \xrightarrow{\sim} V_A.$$

Set

$$x_A := X_{S_A,0} \in X_0 + \mathfrak{u}.$$

Then:

1. The restriction
$$S_A^\vee|_{\ker X_0} : \ker X_0 \to E$$
is surjective, with kernel exactly A.
2. Let $x = X_{C,B} \in X_0 + \mathfrak{u}$, and suppose that
$$\phi_C := C^\vee|_{\ker X_0} : \ker X_0 \to E$$
is surjective. Put
$$L := \ker \phi_C.$$
Assume that
$$L \cap R = 0.$$
Then $q|_L : L \xrightarrow{\sim} q(L) \subset K_{X_0}$ is an isomorphism onto a d-dimensional subspace $A := q(L)$, and x is P-conjugate to x_A.

Moreover, for a fixed A, the P-orbit of x_A is independent of the auxiliary choices of V_A and S_A.

3. If $A_1, A_2 \subset V_1 \simeq K_{X_0}$ are isometric inside K_{X_0}, then x_{A_1} and x_{A_2} are P-conjugate, hence certainly G-conjugate.

4.2 proof

Since $A \subset \ker X_0$ has dimension d, its annihilator in $V_0 / \operatorname{Im} X_0$ has dimension

$$c_1 - d = \kappa,$$

which proves $\dim W_A = \kappa$.

Now let $a \in A$. By definition of W_A, one has

$$\langle a, S_A e' \rangle_0 = 0 \qquad \forall e' \in E',$$

so $S_A^\vee a = 0$. Thus

$$A \subseteq \ker(S_A^\vee|_{\ker X_0}).$$

Conversely, if $k \in \ker X_0$ satisfies $S_A^\vee k = 0$, then k pairs trivially with V_A. Because

$$\ker X_0 = (\operatorname{Im} X_0)^\perp,$$

the value $\langle k, v \rangle_0$ depends only on the class of v in $V_0 / \operatorname{Im} X_0$. Since V_A maps isomorphically onto W_A, the condition $S_A^\vee k = 0$ says exactly that k annihilates W_A under the perfect pairing

$$(V_0 / \operatorname{Im} X_0) \times \ker X_0 \to F.$$

But W_A was defined as the annihilator of A in this perfect pairing, so the double-annihilator property gives

$$k \in \operatorname{Ann}(W_A) = A.$$

Thus the kernel is exactly A. Since

$$\dim \ker X_0 - \dim A = c_1 - d = \kappa = \dim E,$$

the map $S_A^\vee|_{\ker X_0}$ is surjective. This proves (1).

For (2), surjectivity of ϕ_C gives

$$\dim L = c_1 - \kappa = d.$$

The assumption $L \cap R = 0$ implies that $q|_L$ is injective, so

$$A := q(L)$$

is a d-dimensional subspace of K_{X_0}.

We first normalize the restriction of $C^\vee$ to $\ker X_0$. Both maps

$$\phi_C : \ker X_0 \to E \qquad \text{and} \qquad S_A^\vee|_{\ker X_0} : \ker X_0 \to E$$

are surjective and have the same kernel L. Hence there exists a unique $a \in \mathrm{GL}(E)$ such that

$$a \circ \phi_C = S_A^\vee|_{\ker X_0}.$$

Let $d_E \in \mathrm{GL}(E')$ be the unique element satisfying

$$a = (d_E^{-1})^\vee.$$

Conjugating by the corresponding Levi element of P, we may replace (C, B) by a new pair for which already

$$C^\vee|_{\ker X_0} = S_A^\vee|_{\ker X_0}.$$

Then $(C - S_A)^\vee$ vanishes on $\ker X_0$. Equivalently,

$$\langle k, (C - S_A)e' \rangle_0 = 0 \qquad \forall k \in \ker X_0,\ \forall e' \in E'.$$

Since $(\ker X_0)^\perp = \operatorname{Im} X_0$, it follows that

$$(C - S_A)(E') \subseteq \operatorname{Im} X_0.$$

Hence there exists $D_1 : E' \to V_0$ with

$$X_0 D_1 = C - S_A.$$

By Lemma `lem:unipotent-conjugation`, conjugation by u_{D_1} replaces C by S_A.

So we may assume from now on that $x = X_{S_A,B}$. Because $S_A^\vee|_{\ker X_0}$ is surjective, choose $D_2 : E' \to \ker X_0$ such that

$$S_A^\vee D_2 = \frac{1}{2}B.$$

Since $D_2(E') \subset \ker X_0$, one has $X_0 D_2 = 0$, so conjugation by u_{D_2} keeps $C = S_A$ fixed. The new B-block is

$$B' = B + D_2^\vee S_A - S_A^\vee D_2.$$

For $u, v \in E'$,

$$\lambda(D_2^\vee S_A u, v) = -\langle S_A u, D_2 v\rangle_0.$$

Because $S_A^\vee D_2 = \frac{1}{2}B$,

$$\langle D_2 u, S_A v\rangle_0 = -\lambda\left(\frac{1}{2}Bu, v\right).$$

Using $\langle S_A u, D_2 v\rangle_0 = \epsilon\langle D_2 v, S_A u\rangle_0$ and

$$\lambda(Bv, u) = -\epsilon\lambda(Bu, v),$$

we get

$$\lambda(D_2^\vee S_A u, v) = -\frac{1}{2}\lambda(Bu, v).$$

Also

$$\lambda(S_A^\vee D_2 u, v) = \frac{1}{2}\lambda(Bu, v).$$

Therefore

$$\lambda(B'u, v) = 0 \qquad \forall u, v \in E',$$

so $B' = 0$. Thus x is P-conjugate to x_A.

Now fix A and let $\widetilde{V}_A, \widetilde{S}_A$ be another choice with image in W_A, and put

$$\tilde{x}_A := X_{\widetilde{S}_A,0}.$$

By (1), the map

$$\widetilde{S}_A^\vee|_{\ker X_0}$$

is surjective with kernel A. Since

$$A \subset V_1 \qquad \text{and} \qquad V_1 \cap R = 0,$$

we have

$$A \cap R = 0.$$

Therefore Part (2), applied to $\tilde{x}_A$, shows that

$$\tilde{x}_A$$

is P-conjugate to the already chosen x_A. This proves the choice-independence statement.

For (3), let $h : A_1 \to A_2$ be induced by an element of $\operatorname{Isom}(K_{X_0})$. By Lemma `lem:x0-geometry-and-k`, h lifts to an element $\tilde{h} \in Z_{G_0}(X_0)$. Extend $\tilde{h}$ to a Levi element of P as follows. Let

$$W'_{A_2} := \tilde{h}(W_{A_1}) \subset V_0/\operatorname{Im} X_0,$$

choose

$$V'_{A_2} := \tilde{h}(V_{A_1}) \subset V_0,$$

and define

$$S'_{A_2} := \tilde{h} \circ S_{A_1} : E' \xrightarrow{\sim} V'_{A_2}.$$

Then

$$V'_{A_2}$$

maps isomorphically onto W_{A_2}, and the Levi element built from $\tilde{h}$ on V_0, from the induced map on E', and from the dual inverse on E, sends

$$x_{A_1}$$

exactly to

$$X_{S'_{A_2},0}.$$

By the choice-independence already proved, this latter element is P-conjugate to the originally chosen representative x_{A_2}. Hence x_{A_1} and x_{A_2} are P-conjugate.

5 lemma lem:maximal-rank-classes

5.1 statement

Let $\mathscr{A}^{\mathrm{mr}}$ be the maximal-rank locus inside the Grassmannian of d-dimensional subspaces of the real non-degenerate ϵ-space K_{X_0}.

Then the possible isometry classes of elements of $\mathscr{A}^{\mathrm{mr}}$ are finite, and each isometry class is open in $\mathscr{A}^{\mathrm{mr}}$.

More precisely:

1. If $\epsilon = -1$ and d is even, then every element of $\mathscr{A}^{\mathrm{mr}}$ is a non-degenerate symplectic space of dimension d, hence the isometry class is unique.
2. If $\epsilon = -1$ and d is odd, then every element of $\mathscr{A}^{\mathrm{mr}}$ is a symplectic space of rank $d-1$ with one-dimensional radical, hence the isometry class is unique.
3. If $\epsilon = +1$, then every element of $\mathscr{A}^{\mathrm{mr}}$ is a non-degenerate symmetric real bilinear space of dimension d, hence is classified by its signature. In particular only finitely many signatures occur, and each signature class is open.

5.2 proof

If $\epsilon = -1$, the restricted form is alternating. On an even-dimensional space a non-degenerate alternating form is unique up to isometry, and on an odd-dimensional space an alternating form of maximal rank $d-1$ is unique up to isometry: in a suitable basis it is standard symplectic on a $(d-1)$-dimensional subspace and zero on a radical line.

We also need ambient transitivity in these two symplectic cases. If $\epsilon = -1$ and d is even, let $A, B \in \mathscr{A}^{\mathrm{mr}}$. Choose symplectic bases

$$a_1, b_1, \dots, a_{d/2}, b_{d/2}$$

of A and

$$a'_1, b'_1, \dots, a'_{d/2}, b'_{d/2}$$

of B, and extend both to symplectic bases of the ambient space K_{X_0}. The basis-to-basis map is an element of

$$\mathrm{Isom}(K_{X_0}, \langle\cdot,\cdot\rangle_K)$$

carrying A to B.

If $\epsilon = -1$ and d is odd, let r and r' span the radicals of A and B. Choose symplectic bases

$$a_1, b_1, \dots, a_{(d-1)/2}, b_{(d-1)/2}$$

of a complement to Fr in A, and

$$a'_1, b'_1, \dots, a'_{(d-1)/2}, b'_{(d-1)/2}$$

of a complement to Fr' in B. Choose vectors $s, s' \in K_{X_0}$ such that

$$\langle r, s\rangle_K = 1, \qquad \langle r', s'\rangle_K = 1,$$

and extend

$$r, a_1, b_1, \dots, a_{(d-1)/2}, b_{(d-1)/2}, s$$

and

$$r', a'_1, b'_1, \dots, a'_{(d-1)/2}, b'_{(d-1)/2}, s'$$

to symplectic bases of K_{X_0}. Again the basis-to-basis map is an ambient isometry carrying A to B. Thus the ambient isometry orbit is unique in both symplectic cases.

If $\epsilon = +1$, the restricted form is symmetric. Maximal rank then means non-degenerate. Over $\mathbb{R}$, such a form is classified by its signature, and the signature is locally constant on the non-degenerate locus. So only finitely many signatures occur, each open in $\mathscr{A}^{\mathrm{mr}}$.

It remains to see that, for a fixed signature, the corresponding d-planes form a single orbit under $\mathrm{Isom}(K_{X_0}, \langle\cdot,\cdot\rangle_K)$. Let $A, B \subset K_{X_0}$ be non-degenerate d-dimensional subspaces with the same signature. Then there is an isometry

$$u : A \xrightarrow{\sim} B.$$

Because A and B are non-degenerate, the orthogonal complements

$$A^{\perp}, \qquad B^{\perp}$$

are also non-degenerate, and their signatures are the signature of K_{X_0} minus the common signature of A and B. Hence there is an isometry

$$v : A^{\perp} \xrightarrow{\sim} B^{\perp}.$$

Taking orthogonal direct sums gives an ambient isometry

$$u \oplus v \in \mathrm{Isom}(K_{X_0}, \langle \cdot, \cdot \rangle_K)$$

with

$$(u \oplus v)(A) = B.$$

Thus every fixed-signature locus is a single ambient isometry orbit.

6 proposition prop:kernel-image-quotient

6.1 statement

For $A \subset V_1 \simeq K_{X_0}$ of dimension d, let x_A be as in Proposition `prop:generic-normal-form`. Then

$$\ker x_A = E \oplus A,$$

$$\mathrm{Im}\, x_A = E \oplus \mathrm{Im}\, X_0 \oplus S_A(E'),$$

and

$$\ker x_A \cap \mathrm{Im}\, x_A = E \oplus \mathrm{rad}(A).$$

Consequently,

$$\ker x_A / (\ker x_A \cap \mathrm{Im}\, x_A) \simeq A / \mathrm{rad}(A),$$

and the induced form on this quotient is exactly the quotient of $\langle \cdot, \cdot \rangle_K|_A$ by its radical.

In particular, if $A \in \mathscr{A}^{\mathrm{mr}}$ and either $\epsilon = +1$ or d is even, then A is non-degenerate and

$$\ker x_A \cap \mathrm{Im}\, x_A = E, \qquad \ker x_A / (\ker x_A \cap \mathrm{Im}\, x_A) \simeq A.$$

6.2 proof

Take

$$y = e + v + e' \in E \oplus V_0 \oplus E'.$$

Then

$$x_A(y) = S_A^{\vee} v + X_0 v + S_A e'.$$

If $x_A(y) = 0$, then the V_0-component gives

$$X_0 v + S_A e' = 0.$$

The class of $S_A e'$ in $V_0 / \mathrm{Im}\, X_0$ is zero only if $e' = 0$, so $v \in \ker X_0$. Then the E-component gives

$$S_A^{\vee} v = 0,$$

hence $v \in A$ by Proposition `prop:generic-normal-form`(1). So

$$\ker x_A = E \oplus A.$$

The formula for the image is immediate from

$$x_A(e+v+e') = S_A^\vee v + X_0 v + S_A e' :$$

the E-part is $S_A^\vee(\ker X_0) = E$, the $\operatorname{Im} X_0$-part comes from $X_0(V_0)$, and the remaining quotient part comes from $S_A(E')$.

Now let

$$z = e + a \in E \oplus A = \ker x_A.$$

Then $z \in \operatorname{Im} x_A$ if and only if

$$a \in A \cap (\operatorname{Im} X_0 + S_A(E')).$$

We claim that, for $a \in A$, this is equivalent to the class

$$\overline{a} \in V_0/\operatorname{Im} X_0$$

belonging to W_A. Indeed, if

$$a = X_0 v + S_A e',$$

then $\overline{a} = \overline{S_A e'} \in W_A$. Conversely, if $\overline{a} \in W_A$, choose $e' \in E'$ with

$$\overline{S_A e'} = \overline{a};$$

then $a - S_A e' \in \operatorname{Im} X_0$, so

$$a \in \operatorname{Im} X_0 + S_A(E').$$

Now suppose first that

$$a \in A \cap (\operatorname{Im} X_0 + S_A(E')).$$

Then $\overline{a} \in W_A$. For every $b \in A$, the definition of W_A gives

$$0 = \langle b, a \rangle_0.$$

Since $a, b \in V_1 \subset \ker X_0$, this is the same as

$$0 = \langle b, a \rangle_K.$$

Hence $a \in A^\perp \cap A = \operatorname{rad}(A)$.

Conversely, let $a \in \operatorname{rad}(A)$. Then for every $b \in A$,

$$\langle b, a \rangle_K = 0,$$

so again

$$\langle b, a \rangle_0 = 0.$$

Thus the class $\overline{a}$ annihilates A, hence

$$\overline{a} \in W_A.$$

By the claim above this is equivalent to

$$a \in \operatorname{Im} X_0 + S_A(E').$$

Therefore

$$\ker x_A \cap \operatorname{Im} x_A = E \oplus \operatorname{rad}(A).$$

Since E is orthogonal to V_0, the form on $\ker x_A = E \oplus A$ induces exactly the quotient form of $\langle \cdot, \cdot \rangle_K|_A$ on

$$(E \oplus A)/(E \oplus \operatorname{rad}(A)) \simeq A/\operatorname{rad}(A).$$

The final statement is immediate when A is non-degenerate.

7 proposition prop:induced-orbits

7.1 statement

The maximal G-orbits in

$$\mathrm{Ind}_P^G(\mathcal{O}_0)$$

are exactly the G-orbits of the representatives x_A with

$$A \in \mathscr{A}^{\mathrm{mr}},$$

modulo isometry inside $(K_{X_0}, \langle \cdot, \cdot \rangle_K)$.

Equivalently, if $A_1, \dots, A_N$ are representatives for the isometry classes in $\mathscr{A}^{\mathrm{mr}}$, then the maximal induced G-orbits are exactly

$$\mathcal{O}_i := G \cdot x_{A_i} \qquad (1 \leq i \leq N),$$

and these orbits are pairwise distinct.

7.2 proof

Embed G_0 into the Levi factor of P by

$$\iota(g_0)(e + v + e') := e + g_0 v + e'.$$

Then

$$\iota(g_0) X_0 \iota(g_0)^{-1} = g_0 X_0 g_0^{-1}.$$

Also $\mathfrak{u}$ is P-stable: if $p \in P$, then $p(E) = E$ and $p(E^{\perp}) = E^{\perp}$, so the conditions

$$Y(E) = 0, \qquad Y(E^{\perp}) \subseteq E, \qquad Y(V) \subseteq E^{\perp}$$

are preserved by $\mathrm{Ad}(p)$. Therefore

$$\mathcal{O}_0 + \mathfrak{u} = \bigcup_{g_0 \in G_0} \left(\iota(g_0) X_0 \iota(g_0)^{-1} + \mathfrak{u} \right) = \iota(G_0) \cdot (X_0 + \mathfrak{u}),$$

and hence

$$G \cdot (\mathcal{O}_0 + \mathfrak{u}) = G \cdot (X_0 + \mathfrak{u}).$$

So

$$\mathrm{Ind}_P^G(\mathcal{O}_0) = \overline{G \cdot (X_0 + \mathfrak{u})}.$$

We next isolate an open dense subset of the slice. For

$$x = X_{C,B} \in X_0 + \mathfrak{u}$$

define

$$\phi_C := C^{\vee}|_{\ker X_0} : \ker X_0 \to E.$$

Let $\Omega \subset X_0 + \mathfrak{u}$ be the subset of all $X_{C,B}$ such that:

1. ϕ_C is surjective;
2. $L := \ker \phi_C$ satisfies $L \cap R = 0$;
3. the image $A := q(L) \subset K_{X_0}$ lies in $\mathscr{A}^{\mathrm{mr}}$.

This set is open and dense.

Indeed, under the parametrization of Lemma `lem:parametrize-x0-plus-u`, the slice is the product of the affine space
$$\mathrm{Hom}(E', V_0)$$
with the affine space of admissible B-blocks, and the defining conditions for Ω depend only on C.

The assignment
$$\rho : \mathrm{Hom}(E', V_0) \longrightarrow \mathrm{Hom}(\ker X_0, E), \qquad \rho(C) = \phi_C,$$
is linear and surjective. Indeed, for $k \in \ker X_0$ one has
$$\lambda(\phi_C(k), e') = -\langle k, Ce' \rangle_0,$$
so ϕ_C depends only on the class of C in
$$\mathrm{Hom}(E', V_0/\operatorname{Im} X_0).$$
Using the perfect pairing
$$(V_0/\operatorname{Im} X_0) \times \ker X_0 \to F,$$
one gets
$$V_0/\operatorname{Im} X_0 \simeq (\ker X_0)^\vee,$$
and then the pairing $\lambda : E \times E' \to F$ identifies
$$\mathrm{Hom}(E', V_0/\operatorname{Im} X_0) \simeq \mathrm{Hom}(\ker X_0, E).$$
Thus ρ is the quotient map to
$$\mathrm{Hom}(E', V_0/\operatorname{Im} X_0)$$
followed by that isomorphism.

Let
$$\mathrm{Surj} := \{\phi \in \mathrm{Hom}(\ker X_0, E) : \phi \text{ surjective}\}.$$
This is the usual open dense full-rank locus. Put
$$\mathrm{Surj}_R := \{\phi \in \mathrm{Surj} : \phi|_R \text{ injective}\}.$$
Since
$$\ker \phi \cap R = 0 \iff \phi|_R \text{ injective},$$
condition (2) is exactly the requirement that $\phi_C \in \mathrm{Surj}_R$. Because $c_2 = \dim R \le \kappa = \dim E$, injectivity on R is again a maximal-rank condition, hence open and dense. Therefore
$$\rho^{-1}(\mathrm{Surj}_R) \times \{\text{admissible } B\}$$
is an open dense subset of $X_0 + \mathfrak{u}$.

Now define
$$\alpha : \mathrm{Surj}_R \to \mathrm{Gr}_d(K_{X_0}), \qquad \alpha(\phi) = q(\ker \phi).$$
This is well defined because for $\phi \in \mathrm{Surj}_R$ one has
$$\dim \ker \phi = c_1 - \kappa = d$$

and

$$\ker \phi \cap R = 0.$$

We claim that α is a surjective locally trivial map with affine fibers. First fix

$$\phi_0 \in \mathrm{Surj}_R, \qquad L_0 := \ker \phi_0,$$

and choose a complement M_0 to L_0 in $\ker X_0$. On the open neighborhood

$$\mathcal{U}_0 := \{\phi \in \mathrm{Surj}_R : \phi|_{M_0} \text{ is an isomorphism}\}$$

every kernel is the graph of the unique linear map

$$T_\phi := -(\phi|_{M_0})^{-1} \circ \phi|_{L_0} : L_0 \to M_0.$$

Thus

$$\phi \longmapsto \ker \phi$$

is locally trivial on Surj_R.

Next use

$$\ker X_0 = V_1 \oplus R.$$

If $L \subset \ker X_0$ is d-dimensional and satisfies $L \cap R = 0$, then the projection

$$\mathrm{pr}_{V_1} : L \to V_1$$

is injective, hence identifies L with a unique d-plane

$$A_L := \mathrm{pr}_{V_1}(L) \subset V_1$$

and a unique linear map

$$\tau_L : A_L \to R$$

such that

$$L = \{a + \tau_L(a) : a \in A_L\}.$$

Under the isometric identification

$$q|_{V_1} : V_1 \xrightarrow{\sim} K_{X_0},$$

one has

$$q(L) = q(A_L) = A_L.$$

Therefore the map

$$L \longmapsto q(L)$$

from the open subset

$$\{L \in \mathrm{Gr}_d(\ker X_0) : L \cap R = 0\}$$

to $\mathrm{Gr}_d(K_{X_0}) \simeq \mathrm{Gr}_d(V_1)$ is, in the usual graph charts, a trivial affine bundle with fiber $\mathrm{Hom}(A, R)$. Composing this affine bundle with the locally trivial kernel map proves the claim about α.

The map α is surjective: given any

$$A \in \mathrm{Gr}_d(K_{X_0}),$$

view it inside V_1 via $q|_{V_1}^{-1}$ and choose a complement M to A in V_1. Since

$$\dim(M \oplus R) = \kappa = \dim E,$$

there exists an isomorphism

$$\psi : M \oplus R \xrightarrow{\sim} E.$$

Extending ψ by 0 on A gives a surjective map

$$\phi : \ker X_0 = A \oplus M \oplus R \to E$$

with

$$\ker \phi = A \qquad \text{and} \qquad \phi|_R \text{ injective.}$$

Hence $\phi \in \mathrm{Surj}_R$ and $\alpha(\phi) = A$.

Because $\mathscr{A}^{\mathrm{mr}}$ is an open dense subset of

$$\mathrm{Gr}_d(K_{X_0}),$$

its pullback

$$\alpha^{-1}(\mathscr{A}^{\mathrm{mr}})$$

is open dense in Surj_R. Therefore

$$\Omega = \{X_{C,B} : \phi_C \in \alpha^{-1}(\mathscr{A}^{\mathrm{mr}})\}$$

is open dense in $X_0 + \mathfrak{u}$.

We now record the same open dense locus in the more concrete S-language used for the exhaustion argument. Choose a complement

$$V_+ \subset V_0, \qquad V_0 = V_+ \oplus \mathrm{Im}\, X_0.$$

We choose it so that it contains the fixed lift $V_1 \subset \ker X_0$ and

$$V_+ \cap \ker X_0 = V_1.$$

This is possible because $V_1 \cap \mathrm{Im}\, X_0 = 0$. The quotient map identifies V_+ with $V_0/\,\mathrm{Im}\, X_0$, and the perfect pairing

$$(V_0/\,\mathrm{Im}\, X_0) \times \ker X_0 \to F$$

identifies V_+ with $(\ker X_0)^\vee$. For $C : E' \to V_0$, let

$$S_C : E' \to V_+$$

be the composition of C with the quotient map, followed by this fixed identification with V_+. Under the above duality, the map

$$\phi_C = C^\vee|_{\ker X_0}$$

is the transpose of S_C. Hence

$$\phi_C \text{ is surjective} \iff S_C \text{ is injective.}$$

Also

$$V_+ \cap \ker X_0$$

is the annihilator, inside V_+ for this perfect pairing, of

$$R = \ker X_0 \cap \operatorname{Im} X_0$$

so

$$\phi_C|_R \text{ is injective } \iff \overline{S}_C : E' \to V_+/(V_+ \cap \ker X_0) \text{ is surjective.}$$

Here

$$d' := \dim V_+/(V_+ \cap \ker X_0) = \dim R = \dim(\ker X_0^2/\ker X_0) \le \dim E'.$$

Thus the first generic locus may be viewed as the locus of injective maps $S : E' \to V_+$. By Proposition `prop:generic-normal-form`, or equivalently by the explicit Levi and unipotent formulas in Lemma `lem:unipotent-conjugation`, every point on this first locus is P-conjugate to a point of the form

$$X_{S,0} := X_{C_S,0}$$

with $S : E' \to V_+$ injective. Namely, the unipotent action subtracts the $\operatorname{Im} X_0$-part of C, and then a further unipotent with image in $\ker X_0$ kills the B-block.

For such an injective S, put

$$L_S := \{k \in \ker X_0 : \langle k, S(E')\rangle_0 = 0\}.$$

If $\overline{S}$ is surjective, then $L_S \cap R = 0$, and hence

$$A(S) := q(L_S) \subset K_{X_0}$$

is a d-dimensional subspace. The condition

$$A(S) \in \mathscr{A}^{\mathrm{mr}}$$

can be tested, in this S-model, by requiring that the Gram matrix of the pulled-back form

$$(u, v) \longmapsto \langle S(u), S(v)\rangle_0$$

has maximal possible rank among injective S with $\overline{S}$ surjective. Let $\mathcal{U}_2$ be the set of all such $X_{S,0}$.

This set is nonempty: choose a splitting

$$V_+ = (V_+ \cap \ker X_0) \oplus Q$$

with $\dim Q = d'$, map a d'-dimensional summand of E' isomorphically onto Q, and inject the remaining summand of E' into $V_+ \cap \ker X_0$; then choose the injection in the nonempty maximal-Gram-rank locus. The conditions defining $\mathcal{U}_2$ are rank conditions, equivalently the nonvanishing of suitable minors of polynomial Gram matrices. Therefore the P-saturation of $\mathcal{U}_2$ is open dense in the first generic locus. In the notation above, this P-saturation is exactly Ω.

By Proposition `prop:generic-normal-form`, every point of Ω is P-conjugate to some x_A with $A \in \mathscr{A}^{\mathrm{mr}}$. By Proposition `prop:generic-normal-form`(3), isometric subspaces give the same P-orbit. By Lemma `lem:maximal-rank-classes`, only finitely many isometry classes occur, and each class is open in $\mathscr{A}^{\mathrm{mr}}$. Choose representatives $A_1, \dots, A_N$ for these classes, and let

$$\Omega_i := P \cdot x_{A_i}.$$

Then

$$\Omega = \bigsqcup_{i=1}^{N} \Omega_i,$$

and each Ω_i is open in Ω.

Set

$$\mathcal{O}_i := G \cdot x_{A_i}.$$

Then

$$G \cdot \Omega = \bigcup_{i=1}^{N} \mathcal{O}_i.$$

Fix i and write

$$x_i := x_{A_i}.$$

Because

$$\Omega_i = P \cdot x_i$$

is open in the affine slice $X_0 + \mathfrak{u}$, its tangent space at x_i is

$$T_{x_i}(X_0 + \mathfrak{u}) = T_{x_i}(P \cdot x_i) = [\mathfrak{p}, x_i] = \mathfrak{u}.$$

Now consider the map

$$f : G \times (X_0 + \mathfrak{u}) \to \mathfrak{g}, \qquad f(g,z) = gzg^{-1}.$$

Its differential at $(1, x_i)$ is

$$df_{(1,x_i)}(\xi, v) = [\xi, x_i] + v \qquad (\xi \in \mathfrak{g},\ v \in \mathfrak{u}).$$

Since

$$v \in \mathfrak{u} = [\mathfrak{p}, x_i] \subseteq [\mathfrak{g}, x_i],$$

the image of this differential is exactly

$$[\mathfrak{g}, x_i] = T_{x_i}(\mathcal{O}_i).$$

Therefore $G \cdot (X_0 + \mathfrak{u})$ has, near x_i, the same local dimension as the orbit $\mathcal{O}_i$, and the latter is open in $G \cdot (X_0 + \mathfrak{u})$ near x_i. Conjugating by elements of G shows that $\mathcal{O}_i$ is open at each of its points, hence open in

$$G \cdot (X_0 + \mathfrak{u}).$$

Since Ω is dense in $X_0 + \mathfrak{u}$, the subset $G \times \Omega$ is dense in $G \times (X_0 + \mathfrak{u})$, so

$$G \cdot \Omega$$

is dense in $G \cdot (X_0 + \mathfrak{u})$. Therefore each $\mathcal{O}_i$ is an orbit of maximal dimension inside $G \cdot (X_0 + \mathfrak{u})$.

Because

$$\mathrm{Ind}_P^G(\mathcal{O}_0) = \overline{G \cdot (X_0 + \mathfrak{u})} = \overline{G \cdot \Omega},$$

and because

$$G \cdot \Omega = \bigcup_{i=1}^{N} \mathcal{O}_i$$

is a finite union, we get

$$\operatorname{Ind}_P^G(\mathcal{O}_0) = \bigcup_{i=1}^{N} \overline{\mathcal{O}_i}.$$

Thus if $\mathcal{O}$ is any G-orbit in the induced set and

$$x \in \mathcal{O},$$

then

$$x \in \overline{\mathcal{O}_i}$$

for some i. Since $\overline{\mathcal{O}_i}$ is G-stable, the whole orbit $\mathcal{O} = G \cdot x$ is contained in $\overline{\mathcal{O}_i}$. Therefore every orbit in the induced set is below at least one of the listed $\mathcal{O}_i$ in the closure order. It remains to prove that each listed orbit is itself maximal.

Set

$$Y := G \cdot (X_0 + \mathfrak{u}).$$

For fixed i, the slice piece

$$\Omega_i = P \cdot x_i$$

is exactly the set of all $x = X_{C,B} \in X_0 + \mathfrak{u}$ such that:

1. $\phi_C = C^\vee|_{\ker X_0}$ is surjective;
2. $L = \ker \phi_C$ satisfies $L \cap R = 0$;
3. the associated subspace
$$A(x) := q(L) \subset K_{X_0}$$
has maximal possible Gram-matrix rank and lies in the same real isometry class as A_i.

Indeed, conditions (1) and (2) are the defining open conditions for the normalized slice, condition (3) is the intrinsic maximal-rank condition on the kernel quotient parameter $A = q(L)$, and Proposition `prop:generic-normal-form` together with Lemma `lem:maximal-rank-classes` says precisely that the resulting points form the single P-orbit $P \cdot x_i$. Since real isometry class is locally constant on the maximal-rank locus, each Ω_i is open in $X_0 + \mathfrak{u}$. By the differential argument above, the corresponding G-orbit

$$\mathcal{O}_i = G \cdot \Omega_i$$

is therefore open in Y.

Now let $\mathcal{O}$ be a G-orbit in

$$\operatorname{Ind}_P^G(\mathcal{O}_0) = \overline{Y}$$

such that

$$\mathcal{O}_i \subseteq \overline{\mathcal{O}}.$$

Since $\mathcal{O}_i$ is open in Y, it has the same dimension as Y:

$$\dim \mathcal{O}_i = \dim Y.$$

Because Y is dense in

$$\operatorname{Ind}_P^G(\mathcal{O}_0) = \overline{Y},$$

the ambient induced set has the same dimension:

$$\dim \operatorname{Ind}_P^G(\mathcal{O}_0) = \dim Y.$$

Every G-orbit in $\mathfrak{g}$ is a smooth locally closed submanifold, so any orbit contained in the induced set has dimension at most

$$\dim \operatorname{Ind}_P^G(\mathcal{O}_0) = \dim Y.$$

Hence

$$\dim \mathcal{O} \leq \dim Y = \dim \mathcal{O}_i.$$

On the other hand,

$$\mathcal{O}_i \subseteq \overline{\mathcal{O}}$$

forces

$$\dim \mathcal{O}_i \leq \dim \mathcal{O}.$$

Indeed, $\mathcal{O}$ is locally closed, hence open in its closure $\overline{\mathcal{O}}$; therefore every orbit contained in the boundary

$$\overline{\mathcal{O}} \setminus \mathcal{O}$$

has strictly smaller dimension than $\mathcal{O}$.

Combining the two inequalities gives

$$\dim \mathcal{O} = \dim \mathcal{O}_i.$$

If $\mathcal{O} \neq \mathcal{O}_i$, then

$$\mathcal{O}_i \subseteq \overline{\mathcal{O}} \setminus \mathcal{O},$$

so the preceding boundary-dimension statement would imply

$$\dim \mathcal{O}_i < \dim \mathcal{O},$$

contrary to the equality above. Therefore

$$\mathcal{O} = \mathcal{O}_i.$$

So each listed orbit $\mathcal{O}_i$ is maximal.

It remains only to show that the $\mathcal{O}_i$ are pairwise distinct exactly according to the isometry class of A_i. Suppose

$$G \cdot x_A = G \cdot x_{A'}$$

with $A, A' \in \mathscr{A}^{\mathrm{mr}}$. The quotient

$$Q_x := \ker x / (\ker x \cap \operatorname{Im} x)$$

is a G-orbit invariant bilinear space. By Proposition `prop:kernel-image-quotient`,

$$Q_{x_A} \simeq A / \operatorname{rad}(A), \qquad Q_{x_{A'}} \simeq A' / \operatorname{rad}(A').$$

If $\epsilon = +1$, then every element of $\mathscr{A}^{\mathrm{mr}}$ is non-degenerate, so $Q_{x_A} \simeq A$ and $Q_{x_{A'}} \simeq A'$. Hence A and A' are isometric.

If $\epsilon = -1$ and d is even, the same conclusion holds because maximal rank again means non-degenerate.

If $\epsilon = -1$ and d is odd, Lemma `lem:maximal-rank-classes` already says that all elements of $\mathscr{A}^{\mathrm{mr}}$ are isometric. So again the orbit depends only on the isometry class of A.

Therefore the maximal induced G-orbits are parametrized exactly by the isometry classes in $\mathscr{A}^{\mathrm{mr}}$.

8 proposition prop:multiplicity

8.1 statement

For every maximal induced orbit $\mathcal{O} = G \cdot x_A$ with $A \in \mathscr{A}^{\mathrm{mr}}$, one has

$$m(\mathcal{O}, P) = \begin{cases} 1, & \epsilon = +1, \\ 1, & \epsilon = -1 \text{ and } d \text{ even}, \\ 2, & \epsilon = -1 \text{ and } d \text{ odd}. \end{cases}$$

8.2 proof

We treat first the case where A is non-degenerate. Then by Proposition `prop:kernel-image-quotient`,

$$\ker x_A = E \oplus A, \qquad \ker x_A \cap \operatorname{Im} x_A = E.$$

Because A is non-degenerate and E is orthogonal to all of $\ker x_A$, the radical of the bilinear space $\ker x_A$ is exactly E.

Any element of $Z_G(x_A)$ preserves $\ker x_A$, hence preserves its radical. Therefore every element of $Z_G(x_A)$ preserves E, so

$$Z_G(x_A) \subseteq P.$$

Consequently

$$m(G \cdot x_A, P) = 1.$$

It remains to consider the case $\epsilon = -1$ and d odd. Then every $A \in \mathscr{A}^{\mathrm{mr}}$ has one-dimensional radical. Choose a symplectic decomposition

$$A = A_0 \oplus Fr,$$

where A_0 is non-degenerate symplectic and $Fr = \operatorname{rad}(A)$. We now construct a convenient representative for this same A and then invoke the choice-independence statement from Proposition `prop:generic-normal-form`.

Choose

$$V_1 = A_0 \oplus Fr \oplus Fs \oplus B_0$$

with

$$\langle r, s \rangle_K = 1$$

and B_0 symplectic. Choose decompositions

$$E = Fe \oplus E_0, \qquad E' = Fe' \oplus E_0',$$

with

$$\lambda(e, e') = 1$$

and E_0, E_0' paired perfectly. Let

$$W_0 \subset V_0 / \operatorname{Im} X_0$$

be the annihilator of

$$A \oplus Fs.$$

Then

$$\dim W_0 = \kappa - 1.$$

Choose a lift

$$V_0' \subset V_0$$

of W_0 and an isomorphism

$$S_0 : E_0' \xrightarrow{\sim} V_0'.$$

Let $\bar{r}$ denote the class of r in $V_0 / \operatorname{Im} X_0$. Since $r \in \operatorname{rad}(A)$, one has

$$\langle a, r \rangle_K = 0 \qquad \forall a \in A,$$

so

$$\bar{r} \in W_A.$$

On the other hand,

$$\langle r, s \rangle_K = 1,$$

so $\bar{r}$ does not annihilate Fs and therefore

$$\bar{r} \notin W_0.$$

Because

$$\dim W_A = \kappa \qquad \text{and} \qquad \dim W_0 = \kappa - 1,$$

it follows that

$$W_A = W_0 \oplus F\bar{r}.$$

Hence the subspace

$$\widetilde{V}_A := Fr \oplus V_0'$$

maps isomorphically onto W_A.

Define

$$\widetilde{S}_A : E' \xrightarrow{\sim} \widetilde{V}_A$$

by

$$\widetilde{S}_A(e') = r, \qquad \widetilde{S}_A|_{E_0'} = S_0.$$

and put

$$\tilde{x}_A := X_{\widetilde{S}_A, 0}.$$

This is again a representative attached to the same subspace A in Proposition `prop:generic-normal-form`. Therefore $\tilde{x}_A$ is P-conjugate to the original x_A, and

$$m(G \cdot x_A, P) = m(G \cdot \tilde{x}_A, P).$$

Set

$$U := Fe \oplus Fr \oplus Fs \oplus Fe'.$$

Then U is non-degenerate symplectic, $\tilde{x}_A(U) \subseteq U$, and in the ordered basis

$$e,\ r,\ s,\ e'$$

one has

$$\tilde{x}_A(e) = 0, \qquad \tilde{x}_A(r) = 0, \qquad \tilde{x}_A(s) = e, \qquad \tilde{x}_A(e') = r.$$

Because U is non-degenerate and $\tilde{x}_A$ is skew-adjoint, the orthogonal complement

$$W := U^{\perp}$$

is $\tilde{x}_A$-stable. On $U^{\perp}$ the kernel quotient is A_0, which is non-degenerate symplectic. Let

$$y := \tilde{x}_A|_W.$$

The first part of the present proof therefore applies to y: every element of its centralizer preserves the radical of $\ker y$, namely

$$E_0 = E \cap W.$$

Thus

$$Z_{G(W)}(y) \subseteq P(W) := \{h \in G(W) : h(E_0) = E_0\}.$$

We now compute the residual 4×4 model on U correctly. Put

$$U_0 := Fe \oplus Fr, \qquad U_1 := Fs \oplus Fe', \qquad U = U_0 \oplus U_1.$$

Relative to the ordered basis

$$e,\ r,\ s,\ e',$$

the operator $\tilde{x}_A|_U$ is

$$\tilde{x}_A|_U = \begin{pmatrix} 0 & I_2 \\ 0 & 0 \end{pmatrix},$$

and the symplectic form on U is represented by

$$J_U = \begin{pmatrix} 0 & S_2 \\ -S_2 & 0 \end{pmatrix}, \qquad S_2 := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Let

$$G(U) = \mathrm{Sp}(U), \qquad P(U) := \{g \in G(U) : g(Fe) = Fe\}.$$

Write an element of $G(U)$ in 2×2 block form relative to

$$U = U_0 \oplus U_1$$

as

$$g = \begin{pmatrix} A & B \\ C & D \end{pmatrix}.$$

The commutation relation

$$g\tilde{x}_A|_U = \tilde{x}_A|_U g$$

is equivalent to

$$C = 0, \qquad D = A.$$

So every element of $Z_{G(U)}(\tilde{x}_A|_U)$ has the form

$$g = \begin{pmatrix} A & B \\ 0 & A \end{pmatrix}.$$

Now impose the symplectic condition

$$g^T J_U g = J_U.$$

Since

$$g = \begin{pmatrix} A & B \\ 0 & A \end{pmatrix}, \qquad J_U = \begin{pmatrix} 0 & S_2 \\ -S_2 & 0 \end{pmatrix},$$

we first compute

$$J_U g = \begin{pmatrix} 0 & S_2 \\ -S_2 & 0 \end{pmatrix} \begin{pmatrix} A & B \\ 0 & A \end{pmatrix} = \begin{pmatrix} 0 & S_2 A \\ -S_2 A & -S_2 B \end{pmatrix}.$$

Hence

$$g^T J_U g = \begin{pmatrix} A^T & 0 \\ B^T & A^T \end{pmatrix} \begin{pmatrix} 0 & S_2 A \\ -S_2 A & -S_2 B \end{pmatrix} = \begin{pmatrix} 0 & A^T S_2 A \\ -A^T S_2 A & B^T S_2 A - A^T S_2 B \end{pmatrix}.$$

Equating this with

$$J_U = \begin{pmatrix} 0 & S_2 \\ -S_2 & 0 \end{pmatrix}$$

gives

$$A^T S_2 A = S_2$$

and

$$B^T S_2 A = A^T S_2 B.$$

If we write

$$M := A^{-1} B,$$

then the second relation becomes

$$(AM)^T S_2 A = A^T S_2 (AM),$$

that is,

$$M^T A^T S_2 A = A^T S_2 A\, M.$$

Using $A^T S_2 A = S_2$, this is equivalent to

$$M^T S_2 = S_2 M.$$

Thus A lies in the orthogonal group

$$O(S_2) := \{A \in \mathrm{GL}_2(\mathbb{R}) : A^T S_2 A = S_2\},$$

and for each fixed A the possible B form the affine space

$$A \cdot \{M \in M_2(\mathbb{R}) : M^T S_2 = S_2 M\}.$$

Therefore projection to the diagonal block induces a surjective map

$$Z_{G(U)}(\tilde{x}_A|_U) \to O(S_2)$$

with affine fibers that are connected in the usual real topology, and a section

$$A \longmapsto \begin{pmatrix} A & 0 \\ 0 & A \end{pmatrix}.$$

Hence

$$\pi_0(Z_{G(U)}(\tilde{x}_A|_U)) \simeq \pi_0(O(S_2)).$$

Over $\mathbb{R}$, the group $O(S_2)$ admits the explicit parametrization

$$O(S_2) = \left\{ \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} : a \in \mathbb{R}^\times \right\} \sqcup \left\{ \begin{pmatrix} 0 & a \\ a^{-1} & 0 \end{pmatrix} : a \in \mathbb{R}^\times \right\}.$$

Thus there are two diagonal branches and two off-diagonal branches. Each of the two diagonal branches lies in a different connected component according to the sign of a, and the two off-diagonal branches also lies in a different connected component according to the sign of a. Hence there are four connected components in all, so

$$|\pi_0(O(S_2))| = 4,$$

and hence

$$|\pi_0(Z_{G(U)}(\tilde{x}_A|_U))| = 4.$$

If, in addition, g lies in $P(U)$, then g preserves the line $Fe \subset U_0$. Because g acts on U_0 by the block A, this means that A itself preserves Fe. An element of $O(S_2)$ preserves Fe if and only if it has the form

$$\begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} \qquad (a \in \mathbb{R}^\times).$$

So projection to A induces a surjective map

$$Z_{G(U)}(\tilde{x}_A|_U) \cap P(U) \to \mathbb{R}^\times$$

with the same real-topologically connected affine fibers as above and with a section. Consequently

$$|\pi_0(Z_{G(U)}(\tilde{x}_A|_U) \cap P(U))| = 2.$$

Indeed, inside the Fe-stabilizer only the diagonal branches occur, with the two connected components given by

$$a > 0 \qquad \text{and} \qquad a < 0.$$

Finally, write an element of $Z_G(\tilde{x}_A)$, or of $Z_G(\tilde{x}_A) \cap P$, in block form relative to the orthogonal decomposition

$$V = U \oplus W.$$

Since

$$\tilde{x}_A = \tilde{x}_A|_U \oplus y,$$

write

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

where

$$a : U \to U, \qquad b : W \to U, \qquad c : U \to W, \qquad d : W \to W.$$

Projection to the diagonal blocks gives maps

$$Z_G(\tilde{x}_A) \to Z_{G(U)}(\tilde{x}_A|_U) \times Z_{G(W)}(y)$$

and

$$Z_G(\tilde{x}_A) \cap P \to (Z_{G(U)}(\tilde{x}_A|_U) \cap P(U)) \times Z_{G(W)}(y)$$

which are surjective because every pair in either base defines a block-diagonal centralizing element. The second base has the same W-factor as the first because

$$Z_{G(W)}(y) \subseteq P(W).$$

Indeed, relative to

$$E = Fe \oplus E_0$$

with

$$Fe \subset U, \qquad E_0 \subset W,$$

a block-diagonal element

$$\mathrm{diag}(a, d)$$

lies in P exactly when

$$a \in P(U) \qquad \text{and} \qquad d \in P(W).$$

Since every

$$d \in Z_{G(W)}(y)$$

already lies in $P(W)$, intersecting with P imposes no extra condition on the W-factor and only cuts the U-factor down to

$$Z_{G(U)}(\tilde{x}_A|_U) \cap P(U).$$

Fix (a, d) in the base of the first map. The fiber over (a, d) consists of all pairs (b, c) such that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

lies in $Z_G(\tilde{x}_A)$. The commutation relation with $\tilde{x}_A$ is equivalent to the linear equations

$$by = \tilde{x}_A|_U\, b, \qquad c\tilde{x}_A|_U = yc.$$

Let $(-)^*$ denote the adjoint relative to the forms on U and W. Because U and W are orthogonal, the condition $g \in G$ is equivalent to

$$a^*a + c^*c = 1_U, \qquad a^*b + c^*d = 0, \qquad b^*b + d^*d = 1_W.$$

Since $a \in G(U)$ and $d \in G(W)$, this reduces to

$$c^*c = 0, \qquad a^*b + c^*d = 0, \qquad b^*b = 0.$$

Therefore, if (b, c) lies in the fiber, then for every $t \in [0, 1]$ the pair (tb, tc) satisfies the same equations. So each fiber is star-shaped with center $(0, 0)$, hence connected. The same argument applies to the fibers of the second map.

Because both projections are surjective and admit the block-diagonal section, they induce bijections on connected components. Thus

$$\pi_0(Z_G(\tilde{x}_A)) \simeq \pi_0(Z_{G(U)}(\tilde{x}_A|_U) \times Z_{G(W)}(y)),$$

$$\pi_0(Z_G(\tilde{x}_A) \cap P) \simeq \pi_0((Z_{G(U)}(\tilde{x}_A|_U) \cap P(U)) \times Z_{G(W)}(y)).$$

Hence

$$|\pi_0(Z_G(\tilde{x}_A))| = |\pi_0(Z_{G(U)}(\tilde{x}_A|_U))| \cdot |\pi_0(Z_{G(W)}(y))|,$$

$$|\pi_0(Z_G(\tilde{x}_A) \cap P)| = |\pi_0(Z_{G(U)}(\tilde{x}_A|_U) \cap P(U))| \cdot |\pi_0(Z_{G(W)}(y))|.$$

Since

$$|\pi_0(Z_{G(U)}(\tilde{x}_A|_U))| = 4 \qquad \text{and} \qquad |\pi_0(Z_{G(U)}(\tilde{x}_A|_U) \cap P(U))| = 2,$$

the common W-factor cancels and yields

$$m(G \cdot \tilde{x}_A, P) = 2.$$

Because $\tilde{x}_A$ is P-conjugate to x_A, this is exactly

$$m(G \cdot x_A, P) = 2.$$

This completes the case $\epsilon = -1$ and d odd, and hence the proof.

9 theorem thm:main

9.1 statement

10 Short-Kappa Branch of Maximal-Parabolic Induction - Computation Problem

Let

$$F = \mathbb{R}.$$

Let

$$\epsilon \in \{+1, -1\}.$$

10.1 Setup

Let $(V_0, \langle\cdot,\cdot\rangle_0)$ be a finite-dimensional F-vector space with a non-degenerate bilinear form satisfying

$$\langle v, w\rangle_0 = \epsilon\langle w, v\rangle_0 \qquad \forall v, w \in V_0.$$

Set

$$G_0 := \{g \in \mathrm{GL}(V_0) : \langle gv, gw\rangle_0 = \langle v, w\rangle_0\},$$

and

$$\mathfrak{g}_0 := \{Y \in \mathrm{End}_F(V_0) : \langle Yv, w\rangle_0 + \langle v, Yw\rangle_0 = 0 \ \forall v, w \in V_0\}.$$

Let $X_0 \in \mathfrak{g}_0$ be nilpotent, and let

$$\mathcal{O}_0 := G_0 \cdot X_0.$$

Put

$$c_1 := \dim_F \ker X_0, \qquad c_2 := \dim_F(\ker X_0 \cap \operatorname{Im} X_0).$$

Equivalently,

$$c_2 = \dim_F(\ker X_0^2 / \ker X_0).$$

Fix an integer κ, and let E, E' be κ-dimensional F-vector spaces with a non-degenerate pairing

$$\lambda : E \times E' \to F.$$

Assume

$$\dim_F \ker X_0 > \dim_F E = \kappa \geq \dim_F(\ker X_0^2/\ker X_0).$$

Equivalently,

$$c_2 \leq \kappa < c_1.$$

Put

$$d := c_1 - \kappa.$$

Set

$$V := E \oplus V_0 \oplus E',$$

and define a bilinear form on V by

$$\langle e + v + e',\ f + w + f' \rangle := \lambda(e, f') + \epsilon\lambda(f, e') + \langle v, w \rangle_0.$$

Let

$$G := \{g \in \mathrm{GL}(V) : \langle gx, gy \rangle = \langle x, y \rangle\},$$

and

$$\mathfrak{g} := \{Y \in \mathrm{End}_F(V) : \langle Yx, y \rangle + \langle x, Yy \rangle = 0\ \forall x, y \in V\}.$$

Let

$$P := \{g \in G : g(E) = E\}.$$

Since $E^\perp = E \oplus V_0$, define

$$\mathfrak{u} := \{Y \in \mathfrak{g} : Y(E) = 0, \quad Y(E^\perp) \subseteq E, \quad Y(V) \subseteq E^\perp\}.$$

Regard $\mathcal{O}_0$ as a subset of $\mathfrak{g}$ by letting each operator in $\mathcal{O}_0$ act on the V_0-summand and act as 0 on $E \oplus E'$.

Define the induced set

$$\mathrm{Ind}_P^G(\mathcal{O}_0) := \overline{G \cdot (\mathcal{O}_0 + \mathfrak{u})} \subseteq \mathfrak{g},$$

where the closure is taken in the usual real topology on the finite-dimensional real vector space $\mathfrak{g}$.

A G-orbit $\mathcal{O} \subseteq \mathrm{Ind}_P^G(\mathcal{O}_0)$ is called induced from $\mathcal{O}_0$ via P if it is maximal, among the G-orbits contained in $\mathrm{Ind}_P^G(\mathcal{O}_0)$, for the closure order

$$\mathcal{O}_1 \preceq \mathcal{O}_2 \iff \mathcal{O}_1 \subseteq \overline{\mathcal{O}_2}.$$

For $x \in \mathfrak{g}$, write

$$Z_G(x) := \{g \in G : gxg^{-1} = x\}.$$

For a subgroup H of G, let H°_{top} denote its identity component for the usual real topology, and set

$$\pi_0(H) := H/H^\circ_{\mathrm{top}}.$$

For $x \in \mathcal{O}_0 + \mathfrak{u}$, define the multiplicity by

$$m(G \cdot x, P) := \frac{|\pi_0(Z_G(x))|}{|\pi_0(Z_G(x) \cap P)|}.$$

10.2 The kernel quotient

Since X_0 is skew-adjoint, one has

$$(\operatorname{Im} X_0)^\perp = \ker X_0.$$

Therefore the restriction of $\langle \cdot, \cdot \rangle_0$ to $\ker X_0$ has radical

$$\ker X_0 \cap \operatorname{Im} X_0.$$

Define the kernel quotient

$$K_{X_0} := \ker X_0 / (\ker X_0 \cap \operatorname{Im} X_0).$$

The restriction of $\langle \cdot, \cdot \rangle_0$ to $\ker X_0$ descends to a bilinear form on K_{X_0}, denoted by

$$\langle \cdot, \cdot \rangle_K.$$

Its dimension is

$$\dim_F K_{X_0} = c_1 - c_2.$$

The assumption $c_2 \leq \kappa < c_1$ is equivalent to

$$0 < d \leq \dim_F K_{X_0}.$$

Let $\mathscr{A}$ be the set of d-dimensional subspaces

$$A \subset K_{X_0}.$$

For $A \in \mathscr{A}$, write $\operatorname{rad}(A)$ for the radical of the restricted form $\langle \cdot, \cdot \rangle_K|_A$. Let $\mathscr{A}^{\mathrm{mr}} \subseteq \mathscr{A}$ be the maximal-rank locus: this is the set of all A for which the restricted form on A has the largest possible rank among d-dimensional subspaces of K_{X_0}.

Two elements $A_1, A_2 \in \mathscr{A}^{\mathrm{mr}}$ are called isometric inside K_{X_0} if there is an isometry

$$h \in \operatorname{Isom}(K_{X_0}, \langle \cdot, \cdot \rangle_K)$$

such that

$$h(A_1) = A_2.$$

10.3 The slice $X_0 + \mathfrak{u}$

With respect to

$$V = E \oplus V_0 \oplus E',$$

every element of $X_0 + \mathfrak{u}$ has the form

$$X_{C,B} := \begin{bmatrix} 0 & C^\vee & B \\ 0 & X_0 & C \\ 0 & 0 & 0 \end{bmatrix}.$$

Here

$$C : E' \to V_0, \qquad B : E' \to E,$$

and $C^\vee : V_0 \to E$ is determined by

$$\lambda(C^\vee v, a') = -\langle v, Ca' \rangle_0 \qquad \forall v \in V_0,\ a' \in E'.$$

The condition on B is

$$\lambda(Ba', b') + \epsilon \lambda(Bb', a') = 0 \qquad \forall a', b' \in E'.$$

10.4 Problem

Compute the maximal G-orbits in

$$\mathrm{Ind}_P^G(\mathcal{O}_0)$$

and compute their multiplicities with respect to the maximal parabolic subgroup P.

The final answer should give a parametrization of the maximal induced orbits in terms of the isometry classes of the d-dimensional forms obtained from subspaces of

$$K_{X_0} = \ker X_0/(\ker X_0 \cap \mathrm{Im}\, X_0).$$

For each such orbit $\mathcal{O}$, compute

$$m(\mathcal{O}, P) = \frac{|\pi_0(Z_G(x))|}{|\pi_0(Z_G(x) \cap P)|},$$

where $x \in \mathcal{O} \cap (\mathcal{O}_0 + \mathfrak{u})$ is any representative used in the calculation.

Give a self-contained proof from the maximal parabolic action on the slice $X_0 + \mathfrak{u}$. Do not use any diagram parametrization.

10.5 proof

Lemmas `lem:x0-geometry-and-k`, `lem:parametrize-x0-plus-u`, and `lem:unipotent-conjugation` identify the slice and isolate the intrinsic kernel-quotient datum carried by the surjective map

$$\phi_C = C^\vee|_{\ker X_0} : \ker X_0 \to E.$$

Proposition `prop:generic-normal-form` shows that on the open dense locus where ϕ_C is surjective and its kernel is transverse to $R = \ker X_0 \cap \mathrm{Im}\, X_0$, every slice point is P-conjugate to a normal form x_A indexed by a d-dimensional subspace

$$A \subset K_{X_0} \simeq V_1.$$

Among these subspaces, the open dense maximal-rank locus

$$\mathscr{A}^{\mathrm{mr}}$$

is exactly the one that survives as the maximal part of the induced set. Lemma `lem:maximal-rank-classes` shows that only finitely many isometry classes occur there. Proposition `prop:induced-orbits` proves that the corresponding G-orbits

$$G \cdot x_A \qquad (A \in \mathscr{A}^{\mathrm{mr}})$$

are precisely the maximal G-orbits in

$$\mathrm{Ind}_P^G(\mathcal{O}_0),$$

and that two such maximal orbits coincide exactly when the corresponding subspaces are isometric inside $(K_{X_0}, \langle \cdot, \cdot \rangle_K)$.

Finally, Proposition `prop:multiplicity` computes the component-group quotient for every maximal representative and gives

$$m(G\cdot x_A,P)=\begin{cases}1, & \epsilon=+1,\\ 1, & \epsilon=-1 \text{ and } d \text{ even},\\ 2, & \epsilon=-1 \text{ and } d \text{ odd}.\end{cases}$$

Therefore:

1. The maximal G-orbits in $\mathrm{Ind}_P^G(\mathcal{O}_0)$ are parametrized by the isometry classes inside K_{X_0} of the maximal-rank d-dimensional subspaces $A\subset K_{X_0}$.
2. If A is such a subspace and $x_A\in\mathcal{O}_0+\mathfrak{u}$ is the normal representative constructed above, then

   $$\mathcal{O}_A:=G\cdot x_A$$

   is maximal in the induced set, and every maximal induced orbit is obtained in this way.
3. For every maximal induced orbit,

   $$m(\mathcal{O}_A,P)=\begin{cases}1, & \epsilon=+1,\\ 1, & \epsilon=-1 \text{ and } d \text{ even},\\ 2, & \epsilon=-1 \text{ and } d \text{ odd}.\end{cases}$$

This proves the problem.